

1820

Beyond concepts and selling 12... mass production! Frenchman Charles Xavier Thomas de Colmar developed his Arithmometer, the first mass-produced calculator. Its multiplicative operations followed the same general principle as Leibniz's machine. This was the most *reliable* calculator developed thus far, and that was probably why it was mass-produced. This, and other machines with the same overall design, continued to be sold right up to around 1910.

1.3 Babbage

Who was the Father of Computing? There are several popular answers, depending on context; the favourites seem to be, in this order:

1. Alan Turing, who laid down certain rules regarding what a computing machine is
2. Charles Babbage, who designed what we'll describe now
3. The inventors of the transistor (arguably the greatest invention of the 20th century and the 21st century thus far): John Bardeen, Walter Brattain, and William Shockley
4. Jack Kilby of Integrated Circuit fame

But before we begin about Babbage, here's what a "difference engine" is, from Wikipedia (handle with care): *A difference engine is a special-purpose mechanical digital calculator designed to tabulate polynomial functions. Logarithmic and trigonometric functions can be approximated by polynomials.*

If you don't know what these are, suffice it to say that operations on polynomials, trigonometric functions, and such are much more advanced than arithmetic operations. In addition, the first computers were, like today, number-crunchers (though they didn't crunch much); the point is that ~direct mathematical number crunching~ was what they were destined for, unlike, say, the number-crunching that a graphics card does (because it's ultimately for the sake of pixels). In sum, the early computers just operated on numbers, period.